

PURCHASE DESCRIPTION FOR THE ACQUISITION OF A Wire EDM Machine

1.0 SCOPE.

1.1 Scope. The content of this specification describes the minimum requirements for the replacement of one (1) Wire EDM machine. This shall include the removal of the old wire EDM and the installation of the new Wire EDM. The machine shall include all equipment, accessories, attachments, delivery, installation services, and training described herein.

2.0 APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.2 Government Documents.

2.2.1 Specifications, Standards, and Handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

STANDARDS – FEDERAL

FED-STD-H28B: Screw Thread Standards for Federal Service

U.S Department of Labor, Occupational Safety and Health Administration (OSHA)

29 CFR 1910 Vol. 5: General Industry, OSHA Safety and Health Standards

(Application for copies should be addressed to the OSHA Publications Superintendent of Documents P.O. Box 37535 Washington, D.C. 20013-7535)

DEPARTMENT OF DEFENSE

MILITARY STANDARDS

MIL-STD-130N w/Change 1: Identification Marking of U.S. Military Property Item Unique Identification Marking (IUID)

(Application for copies should be addressed to Air Force Product Data Systems Modernization Standards (TMSS) Office, 554 ELSG/SBP, 4375 Chidlaw Road, Bldg. 262, Room S008, Wright-Patterson AFB, OH 45433-5006, Phone (937) 257-0870 or 0853 or <https://quicksearch.dla.mil/qsSearch.aspx>)

2.2.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

LOCAL FACILITY DOCUMENTS

RESERVED

(Applications for these documents should be addressed to the contracting officer)

2.3 Non-Government Publications, Specifications, Standards, and Handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

American National Standards Institute, Inc. (ANSI)

B11/STD B11: General Safety Requirements Common to ANSI B11 Machines

B11/STD B11.27: Safety Requirements for EDM equipment and EDM systems.

(Address application for copies to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)

International Organization for Standardization (ISO)

ISO 14137: Test conditions for electrical-discharge-machines (wire EDM) test and accuracy.

ISO 4413: Hydraulic fluid power-General rules and safety requirements for systems and their components

ISO 4414: Pneumatic power-General rules and safety requirements for systems and their components

(Application for copies should be addressed to the International Organization for Standardization (ISO) 1 rue de Varembé, Case postale 56, CH211 Geneva 20, Switzerland)

National Electrical Manufacturers Association (NEMA)

NEMA 250: Enclosures for Electrical Equipment (1000 V Maximum)

NEMA ICS 1: Industrial Control and Systems General Requirements

NEMA ICS 6: Industrial Control and Systems: Enclosures

NEMA ICS 16: Industrial Control and Systems: Motion/Position Control Motors, Controls, and Feedback Devices

NEMA MG 1: Motors and Generators

NEMA Z535.4: Product Safety Signs and Labels

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Rosslyn, VA 22209)

National Fire Protection Association (NFPA)

NFPA 70: National Electrical Code

NFPA 79: Electrical Standard for Industrial Machinery

Society of Automotive Engineers (SAE)

SAE EIA-267-C: Axis and Motion Nomenclature for Numerically Controlled Machines

SAE EIA-274-D: Interchangeable variable block data contouring, format for positioning and contouring/positioning numerically controlled machines

(Application for copies should be addressed to: Society of Automotive Engineers 400 Commonwealth Drive, Warrendale PA 15096)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS.

3.1 Design. The item offered shall be new and a current commercially available model. A current model is defined as the manufacturer's currently produced model which, on the date this solicitation is issued, has been designed, engineered, sold, or is being offered for sale through advertisements or manufacturer's published catalogue or brochures. Products such as a prototype unit, pre-production model, or experimental unit do not qualify as meeting the requirements specified herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measuring and Indicating Device Graduations. All measuring and indicating devices on the machine shall be graduated in the US Customary Units.

3.1.2 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's workstation(s).

3.1.3 OSHA Compliance. The equipment installation and its component parts shall be in compliance with 29 CFR 1910, NFPA 70 and NFPA 79. Approval shall be as specified under the "Approval" and "Acceptance" criteria in the OSHA regulations Subpart "O", Machinery and Machine Guarding paragraph 1910.212 and Subpart "S" Electrical, paragraph 1910.303 and paragraph 1910.399.

3.1.4 Safety and Health Requirements. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether specifically required herein, shall comply with all the requirements of OSHA 29 CFR 1910. Safety signs and labels on all equipment shall comply with NEMA Z535.4.

3.1.5 Safety Guarding. Covers, guards, or other safety devices shall be provided. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be easily removed to facilitate inspection, maintenance, and repair of the parts.

3.1.6 Audible Noise Level. Audible noise emitted by the equipment shall not exceed 84 decibels (dB), measured on the "A" weighted scale of a standard Type II sound level meter, at the operator's work position or any point at three (3) feet from the equipment. Noise generated by the work piece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

3.1.7 Mercury Restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture with the exception of fluorescent lighting where required.

3.1.8 Asbestos Restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.1.9 Lead. Lead and materials containing lead shall not be used on or in the machine with the exception to radiation cladding and lead solder in circuits.

3.1.10 Use of Polychlorinated Biphenyl (PCB). The use of polychlorinated biphenyl (PCB) on or in the equipment is prohibited. The machine shall be considered PCB free providing certification to be verified containing less than 5 parts per million (<5ppm) in accordance with 40 CFR part 761.

3.1.11 Environmental Protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.1.12 Lubrication. The machine shall be equipped with a system that provides lubrication (oil, grease, coolant, etc.) for all moving parts in accordance with manufacturer requirements. Recirculating lubrication systems shall include a filter that is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity. Means shall be provided to indicate a low lubrication condition. When a low lubrication condition occurs, the machine shall complete the current operation and come to a complete stop. The machine shall indicate that lubrication is required. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.1.13 Replacement Parts. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Replacement parts are those that are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include but not be limited to motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.2 Construction. The machine shall be constructed of parts that are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 Castings and Forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 Fastening Devices. All screws, pins, bolts, and other fasteners shall be installed to prevent unintentional loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.2.3 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28B and the applicable "Detailed Standard" section referenced therein.

3.2.4 Gears. All gears of axis drive trains shall conform to or exceed all provisions of ANSI/AGMA ISO 1328-1-B14. All gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in drive trains shall be hardened and ground steel.

3.2.5 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.6 Welding, Brazing, or Soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.7 Painting. The machine shall be painted in accordance with the manufacturer's commercial practice. All paint shall be of the lead-free type.

3.2.8 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this purchase description.

3.3 Components. The machine shall consist of not less than the following components:

3.3.1 Base. The base shall be a stress relieved, heavy duty, ribbed casting having characteristics that minimize both thermal and vibration effects, while providing the strength and size to assure maximum rigidity and support for any load within the rated capacity of the machine. Means shall be provided for accurate, ease of leveling.

3.3.2 Table. A stainless-steel fixed worktable shall support and withstand the maximum capacity work-piece loads. The worktable shall be designed to allow the upper and lower wire guide to pass through without interference. An Integrated clamping system shall be provided for the worktable. The worktable shall be provided with threaded holes for securing the work piece utilizing the integrated clamping system. The top of the worktable shall be machine finished for a work mounting surface and parallel to the X and Y axes to maintain the positioning accuracy specified. The worktable shall have a trough or similar means for draining dielectric fluid into its reservoir. The machine shall include a door for access to the cutting area and be arranged to allow for overhead loading of worktable.

3.3.2.1 Indexing Rotary Table. An indexing and simultaneously rotating table shall be provided as an accessory to the work table. Any ancillary equipment needed to connect and operate correctly shall be provided.

3.3.3 Wire Guide Column. A wire guide column shall be designed and constructed as to provide adequate support for the wire, wire tensioning, feed mechanism, wire guides, and mounted servo control motion components. The column shall be securely attached to the base and have provisions for vertical and horizontal travel of the upper wire guide within the specified range. A wire spool holder shall be provided to support a minimum of 55lb. spool of wire.

3.3.4 Wire Path Components. Upper and lower closed style high precision diamond wire guides shall be provided in order to maintain the wire tension and position accuracy. Simultaneous motion control of both upper and lower guide assemblies shall facilitate the machining of angles and conical with a maximum angle as specified in 3.4 table 1. Automatic wire alignment, positioning, measurement of distance between guides and lateral/adjustable flushing nozzles shall be provided. Closed style high precision diamond wire guides shall be provided to allow use of 0.006", 0.008", 0.010", and 0.012" wire. Wire shall be extracted from the lower head and deliver the spent wire to a chopper. The chopped wire shall be deposited into a removable container that shall hold a minimum of 55lb. roll of wire before requiring to be emptied.

3.3.5 Dielectric System. The dielectric system shall supply an adequate amount of dielectric fluid to the cutting area to maintain stable operating temperature and conductivity

levels to efficiently run at the highest cutting rate and surface finish. Control over the dielectric shall be both manual and automatic. The system shall provide a means for recirculation of the dielectric fluid and have a self-contained filtration system, which is easily cleaned and serviced. The filters shall remove all particulates down to Five (5) μm (microns). The dielectric system shall be a close loop, completely assembled unit that is easily accessible for maintenance purposes. There shall be means to contain the spillage of dielectric fluid at the cutting area.

3.3.6 Hydraulic System. The hydraulic system and all components shall conform to ISO 4413. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls.

3.3.7 Pneumatic System. The pneumatic system and all components shall conform to ISO 4414. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls.

3.3.8 Electrical System. The machine shall conform to NFPA 79. The existing source power available at the facility is 460 volt, 3-phase, 60 Hertz (Hz). The machine's electrical system shall be tolerant of fluctuation of ± 10 percent. The electrical system shall be complete, including any electrical transformer(s) that may be required to modify the existing source voltage to the proper operating voltage of the equipment. A properly rated, and fused, single disconnect device shall be utilized on the machine with means of lockout in the off position only.

3.3.8.1 Electrical Enclosures. Construction of the electrical enclosures shall conform to NEMA ICS 6 standards. The enclosure shall be a NEMA type 12 and shall provide for ventilation of components and shall be designed for an indoor non-hazardous location. All electrical components shall conform to applicable NEMA ICS 1 standards. The machine shall be provided with interlocks on its electronic cabinet that shall instantly remove power from the equipment when the door is opened in accordance with NFPA 79.

3.3.8.2 Arc Flash Protection. The machine enclosures shall be marked for arc flash in accordance with NFPA 79 Paragraph 16.2.3 "Safety Signs for Electrical Enclosures" for arc flash safety.

3.3.8.3 Solid-State Components. Electrical devices used to power electrical servo and spindle motors of control voltage delivered to solid-state electrical components shall be solid-state devices. Such devices shall handle the electrical load imposed by the operation of the machine to its maximum capability. The use of selenium in solid-state components shall be restricted to those devices that protect other solid-state components from voltage surge.

3.3.9 Controller Features. The machine controller shall consist of a CNC Control incorporating all standard features. The CNC unit shall provide programmable automatic control of machine functions, operating modes, axis movement and other part program directed functions for the machine. Movement in all axes shall be both synchronous and independent positioning. Control features shall include Manual Data Input (MDI), simultaneous control of all axes, linear interpolation, circular interpolation, programmable

interface, part program edit, buffer edit capability, part program storage, fixed cycles, and control diagnostics. The CNC shall direct machine functions from command data stored in memory. The data stored in memory shall be input by the media specified herein. The control shall initiate a halt and error signal should a fault condition occur in the control. The control shall automatically shut down when internal operating temperatures exceed safe operating limits. The controller shall also provide, but not limited to the following features:

- A. Collision protection
- B. Tri-color warning lights
- C. Color monitor
- D. Glass scales
- E. 3-D automatic part pickup and skewing for part alignment and level compensation.
- F. Automatic and manual wire threading, rethreading capability.
- G. Automatic finding for: corner, center of hole, center of slot, center of slot edge.
- H. Automatic wire offset.
- I. Automatic power restart.
- J. Battery backup.
- K. Display of wire center path along a two (2) or 3-D path.
- L. Dry run movement.
- M. Emergency stop.
- N. Inch/metric switchable data selection.
- O. Incremental/absolute programing.
- P. Linear and circular interpolations.
- Q. Mirror Images X and Y.
- R. Part scaling.
- S. Part rotation.
- T. Reference point return.
- U. Decimal point programing.
- V. Software functions shall be menu driven with interactive conversational prompting to allow for programing at the machine control unit through QWERTY keyboard.
- W. Parametric programing.
- X. Feed rate Programing – IPM/MMPM
- Y. Program edit, change add, and block delete.
- Z. Reference zero (all axes position)
- AA. Key lock for programing edits.
- BB. Wire break alarm.

3.3.9.1 Remote Pendant. A handheld pendant control shall be provided which provides manual positioning control of all axes at selectable feed rates. The pendant control shall be connected to the machine in a method which enables the operator to control the machine functions while in close observation position. The control pendant shall include but not limited to the following:

- A. Machine stop.
- B. Axes Jog (all axes).

- C. Axes positioning speed
- D. Positioning by means of an electronic hand wheel.
- E. Emergency stop.

3.3.9.2 Universal Serial Bus (USB) Connection Port. A minimum of one (1) Standard USB connection port for the transfer of process information to the system control shall be provided. All protocol documentation shall be supplied.

3.3.9.3 Ethernet Connection Port. An RJ-45 connection port for the Ethernet shall be readily available on the exterior of the machine. The CNC controls shall have the capability to upload and download part program results using standard Ethernet protocol.

3.3.9.4 CNC and Control Restoration Data. The contractor shall be responsible for the generation of all programming logic and documentation necessary for machine operation. The Programmable Memory Controller (PMC) program, CNC parameters and all data required for configuring a new control and any other configured or programmed electronic device shall be provided in electronic format on read-only optical disc and delivered with the equipment. For safety and security reasons, the vendor shall not provide encrypted files containing proprietary cycles PMC, PLC programs/data.

3.3.9.5 Software Package. A post processor for use with machine control compatible with ESPRIT CAM most recent software shall be provided. Third party software shall not be used to develop the post processor nor required to operate the post processor. The post processor shall include all necessary Virtual Machine Manager (VMM) files, Machine Simulation Models, and Machine Definition Documents (MDD) to provide the capability to generate tool paths and output ASCII format NC files for all available functionality of the machine including but not limited to, simultaneous 2, 3, 4, and 5 axis machining. The files shall include all documentation. Testing and Debugging of the post processor shall be the sole responsibility of the contractor. Any and all damage from testing/prove out of the post processor shall be the sole responsibility of the contractor and all repairs shall be made by the contractor at their expense. The functioning post processor shall become part of the final acceptance testing and witnessed by the Government. The final acceptance shall not be completed until a fully functioning post processor is demonstrated.

3.3.9.6 Software. Machine shall have ESPRIT Wire EDM software. The vendor shall provide the license to this software.

3.3.9.7 Hardware. Laptop Requirements

- A. Dell Pro Max 18 Plus
- B. Windows 11 Pro English 23H2
- C. Processor: 32-Core 2.2 GHz 13th Gen Intel Core i9-13950HX Intel64 Family 6 Model 183 Stepping 1 Processor speed 2200 (Min)
- D. Memory: 64GB, 1x32GB NECC 4800MHz DDR5 CAMM Module (min)
- E. Video Graphics Card: NVIDIA RTX 5000 Ada Generation Laptop GPU (min)
- F. Hard Drive: M.2 2280 1TB, Gen 4 PCIe x4 NVMe, Solid State Drive
- G. Optical Readers shall be external
- H. Intel Wi-Fi 6E AX210 Wireless Card with Bluetooth 5.2

- I. Wireless Driver: Intel Wi-Fi 6E AX211 160MHz
- J. Intel Ethernet Connection (22) I219-LM
- K. TPM Module (Trusted Platform Module 2.0)
- L. 110 Volt Power Cord
- M. 1 Year onsite service with 1 Year NBD Limited Onsite Service After Remote Diagnosis
- N. Palm Rest, with Smart card only (Microsoft Usbccid Smartcard Reader (WUDF))
- O. 17.3" Anti Glare Laptop Screen with Camera and Mic
- P. CAT 6 cable connection to equipment, computer, and facility local area network, all 4 connections simultaneously active.
- Q. Shall be TAA compliant.
- R. MAC address for laptop shall be provided to customer at least 60 days prior to delivery.

Accessories for Laptop

- A. Two (2) Dell UltraSharp 24 InfinityEdge Monitors - U2422H
- B. Dell MDS19 Dual Monitor Stand
- C. 240W AC Adapter
- D. Dell Smartcard Keyboard KB813/ and corded mouse.
- E. Dell Performance Dock - WD19DC 240W PD

3.3.10 Motors. Motors shall be rated for continuous duty. Alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1. Each motor shall have an identification plate in accordance with paragraph 3.7 to identify manufacturer, model number, serial number, voltage, amperage, horsepower, phase, frequency, and frame number. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's design and construction drawing(s).

3.3.11 Work light. A work light, whether built-in or added on, shall be provided. The light shall comply with NFPA 79 requirements, have a protective shield, and provide maximum light to the work area.

3.3.12 Facility Connections. Electrical connections made to the facility electrical system shall conform to NFPA 70.

3.3.13 Hour Meter. The machine shall be equipped with a non-resetting type, analog hour meter to display accumulated operating time of the drive motor(s), mounted on the main electrical panel visible from the outside, sealed to prevent the entrance of dust and moisture, and shall be mounted to withstand shock and vibration generated by the machine. A CNC control that tracks and displays total machine on time (a non-resetting type), controller on, program runtime, and run time of the servo motor(s) run time shall be considered to meet this requirement. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4 Size and Capacity. The size and capacity of the machine shall be not less than specified in Table I.

Table I Size and Capacity	
Machine Max Dimensions	204 inches x 240 inches x 120 inches
Workpiece Max Dimensions (W, D, H)	48 Inches x 30 Inches x 20 Inches
Workpiece Weight	6000 lbs.
Table Dimensions	46 Inches x 28 Inches
Machining Range (XYZ)	30 Inches x 20 Inches x 20 Inches
U/V Travel (from center)	30 Inches x 20 Inches
Taper Angle at Max Hight	48°
Solid Round Diamond Guides	Closed diamond type without clearance
Resolution Glass Scales	0.000002 Inches
Wire Diameter Range (Inches)	0.006, 0.008, 0.010 and 0.012 Inches (.010 Inches Standard)
Minimum Wire Thread Start Hole	0.020 Inch
Maximum Workpiece AWT Height	16.0 Inches
Wire Spool	55 lbs.

3.5 Standard Tooling and Equipment. All standard tooling and equipment required for proper operation of the machine (such as special tools needed for maintenance) shall be furnished. A list of these items shall be furnished at time of proposal.

3.5.1 Required Accessories. All standard tooling and equipment required for proper operation of the machine (such as special tools needed for maintenance) shall be furnished. A list of these items shall be furnished at time of proposal. The following shall also be included as a minimum:

- a. 2 sets wire guides for .006, 0.008, .010 and .012 inches size wire

3.6 Marking on Instruments, Control Panels, and Plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.6.1 Lubrication Plate. A lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of Lubricant Application
 Servicing Interval
 Type of Lubricant
 Viscosity
 Military or Federal Specification for Each Lubricant (If available)
 Size and Type of Each Lubricant Filter

3.6.2 Nameplate. A nameplate shall be permanently and securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature

Manufacturer's Name

Manufacturer's Model Designation

Manufacturer's Serial Number

Power Input (Volts, Total Amps, Phase, Frequency)

Contract Number

Date of Manufacture

Full-load current for each supply

Ampere rating of the largest motor or load

Short-circuit current rating of the industrial control panel

Max ampere rating of short-circuit and ground fault protective device, where provided

Electrical diagram number(s) or number of the index to the electrical drawings

3.6.3 Identification Marking of Military Property. An Item Unique Identification Marking (IUID) shall be provided in accordance with MIL-STD-130N and all applicable documents within the standard with Machine Readable Information (MRI) for item identification marking and automatic data capture. The application of Human Readable Information (HRI) shall be used in combination with MRI and free text. At a minimum this tag shall contain the information listed in paragraph 3.7.2 Nameplate.

3.7.4 Technical Data. Two (2) hard copies and three (3) electronic copies of technical data shall be provided with the machine. Each electronic copy shall be provided in pdf. or a read-only optical disc. All copies of technical data shall be legible and written in the English language. Technical Data shall include the following

- a. All engineering drawings shall be in accordance with ASME Y14.100. Technical data shall contain technical information, specifications, instructions for the operation, maintenance of the machine, consumables and expendables information (such as oils, fluids, grease, filters...), and recommended spare parts list.
- b. Factory service and operator manuals for the type and model of the machine being purchased.
- c. Diagnostic procedures for isolating and correcting faults. Procedures shall be included in the service manuals for recalibrating and performing operational checks to ensure proper operation of any replaced parts, as required.
- d. Preventative and predictive maintenance tasks, including oil and vibration analysis, and show recommended intervals of tasks.
- e. Software manuals and registration keys/codes shall be provided, along with original software installation disks.
- f. The Contractor shall provide the Government a copy of all test results completed in section 4.7 of this specification.

3.6.4 Plan of Action and Milestones (POA&M). The contractor shall provide a POA&M to identify contractor performance dates of all major events from contract award to completion of all contract requirements. The initial POA&M shall be delivered thirty (30) calendar days After Contract Award (ACA). The milestones shall be prepared using a recognized project management technique (preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M shall be approved in advance by the Government. The POA&M shall detail such items as system design concept, system development, software design concept, software development, engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

3.6.5 Control of Hazardous Energy (Lockout/Tagout) Procedures. A written procedure for isolation of all energy sources on all equipment provided under this contract, in accordance with 29 CFR 1910.147, shall be provided by the contractor and included in the technical manuals. The draft procedures shall be submitted in Microsoft Word compatible format thirty (30) calendar days ACA for Government review and approval.

3.6.6 Instrument and Calibration Procedures. Calibration procedures and/or tooling requiring operator verification or recalibration as part of a production run or on a daily or periodic basis shall be provided to the Government. The contractor shall provide a list of these items at the time of proposal. All items shall have a certificate of calibration traceable to the National Institute of Standards and Technology (NIST). The contractor shall submit draft procedures thirty (30) calendar days ACA for Government review and approval.

3.6.7 Acceptance Test Plan (ATP). The Contractor shall prepare an Acceptance Test Plan (ATP). The ATP shall be a step-by-step written instruction on how to verify compliance during Contract Quality Assurance Inspection and the Final Inspection/Acceptance Test specified in Sections 4.4 and 4.5. **The Contractor shall provide dimensioned drawings of the proposed test part specified in Paragraph 4.7.3.** The Manufacturer's standard testing procedures shall be included in the ATP. The ATP shall include any procedures necessary for compliance with Sections 4.4 and 4.5 that are not covered by the Manufacturer's standard testing procedures. The initial ATP shall be submitted thirty (30) calendar days ACA for Government review and approval. The approved ATP shall become part of the contract.

3.6.8 Part Model and Drawings. The Contractor shall develop a test part model (DXF) for the performance testing outlined in section 4.7.3. The Contractor shall submit the test part model and drawing along with the POA&M. The test part shall utilize all functions and features of the machine. The Government will review the test part model within fifteen (15) business days of submission. The Contractor shall re-submit any required changes to the test part model within five (5) business days for review and approval by the Government. The Contractor shall provide the Government dimensioned drawings of the approved test part thirty (30) calendar days prior to Origin testing.

3.6.9 Equipment Drawings. The Contractors proposal shall be provided with dimensioned equipment drawings in both a .PDF file and CAD file compatible with AutoCAD 2015 edition. The dimensioned CAD file shall be drawn to actual size and show footprint and any manufacturer's special foundation requirements of new equipment to include ancillary pieces, utility connection points, access doors and required maintenance clearance dimensions. The CAD file shall show side and elevation views.

3.7 Installation. Contractor shall be responsible for all equipment (forklifts, man lifts, etc.) and qualified personnel required for unloading, rigging, installation, anchoring, connecting machine to utilities, and leveling the machine in place. The Contractor shall provide the necessary anchor bolts, leveling screws and related hardware to secure machine components to a foundation. Contractor shall also be responsible for the Installation Responsibilities sheet. Contractor shall be responsible for reading, understanding, and complying with Government facility requirements.

3.7.1 Field Supervisor. The Contractor shall provide a full-time field supervisor to direct installation and testing of the supplies from start of installation to final acceptance. The field supervisor shall be capable of reading, writing, and conversing fluently in the English language. The field supervisor shall have full authority to implement field decisions expeditiously. No work shall be accomplished when the field supervisor is not in the immediate work area. The field supervisor shall be identified to the receiving activity prior to start of installation. The field supervisor shall be responsible for coordinating with the receiving activity for any security clearances for all contractor personnel expected to perform on-site work. The receiving activity Government Point of Contact shall be notified of any change in personnel performing this duty.

3.7.2 Installation Plan. The Contractor shall be responsible for providing an Installation Plan. The Installation Plan shall contain detailed information regarding all installation work. The initial Installation Plan shall be delivered to the Government within thirty (30) calendar days After Contract Award (ACA) for review and approval.

3.7.3 Compressed Air Connection. The Contractor shall be responsible for design and installation of new compressed air piping and associated components. The existing compressed air pressure available at the Government facility fluctuates between 83 to 93 psi. The contractor shall assume that all compressed air qualities to be of the lowest quality that is useable as compressed air. The Contractor shall make provisions to filter, dry, humidify, lubricate, amplify, and regulate the Government supplied compressed air to meet the Original Equipment Manufacturer (OEM) compressed air requirements and recommendations.

3.7.4 Existing Equipment and Utilities Removal. The contractor shall be responsible for removing replaced equipment and utilities. The existing equipment is an ONA wire EDM (asset: 65923083487) located in Building 4225. The removal of utilities shall include, but not be limited to, associated components, piping, conduit, and wiring. The Contractor shall transport the equipment to Building 4466 for decontamination and disposal by the Government. Estimated travel is two (2) miles within the confines of the facility. The Contractor shall be responsible for on-site disassembly of machine. All disassembled pieces shall be less than 8 feet by 8 feet by 40 feet and weigh less than 40,000 pounds. The

Government will decontaminate the machine on-site upon completion of disassembly by Contractor. The Contractor shall transport the equipment to DLA Disposition Services upon completion of decontamination by Government. Estimated travel is two (2) miles within the confines of the facility.

3.7.5 Machine Foundation. The Contractor shall be responsible for providing a machine foundation conforming to Original Equipment Manufacturer (OEM) requirements. For unforeseen issues that are found during installation of the foundation, the contractor shall notify the facility point of contact (POC) and the Contracting Officer immediately to determine further course of action.

3.7.6 Telecommunications Connection. The Contractor shall be responsible for providing connection to the Government facility Distributed Numerical Control (DNC) system. Four (4) RJ45 ports shall be provided, installed, and simultaneously active. Estimated distance is 50 feet and height of ceiling is 20 feet.

3.8 Training. After satisfactory completion of acceptance testing of the system, the services of a qualified representative(s), who is an American citizen of good standing and repute and proficient in the English language, shall be provided for specialized training to familiarize Government personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The course shall utilize as-built drawings and manuals provided under this contract and shall include both classroom and hands on portions. All printed training aids shall be supplied for the course, be in the English language, and become the property of the Government.

3.8.1 Operator Training. Operator training shall be provided at FRC East. Training shall be provided for a minimum of three (3) operators. Training shall not be less than three (3) consecutive eight (8) hour workdays, between 6:30 am and 3:00 pm excluding weekends and federal holidays. Training shall include preparation of equipment for safe operation and proper isolation of equipment procedures. The operator shall be provided instruction for the care, use and operation of all attachments and accessories. Training manuals shall be provided for each student.

3.8.2 Maintenance Training. Mechanical and Electronic/Electrical maintenance training shall be provided for four (4) maintenance personnel each. Training shall be performed at the Government facility and shall not be less than three (3) consecutive eight (8) hour workdays each, between 6:30 am and 3:00 pm excluding weekends and federal holidays. Training manuals shall be provided for each student. The maintenance course shall cover at a minimum:

- a. review of systems drawings and schematics
- b. component location and function
- c. troubleshooting procedures and techniques
- d. repair procedures including disassembly and assembly
- e. system maintenance and servicing
- f. setup, adjustments, and calibration (how, when, and where)
- g. preventative maintenance procedures

- h. Procedures for proper isolation.
- i. Basic start up, manual, auto, and MDI mode
- j. SRAM, Ladder, PMC Memory, Parameters, Backup and Restoring

3.8.3 OEM Facility Maintenance Training. Mechanical and Electronic/Electrical maintenance training shall be provided for six (6) total maintenance personnel. The training location shall be performed at a certified training facility of the machines manufacturer and shall not be less than thirty-two (32) hours ie: four (4) consecutive eight (8) hour workdays. Training shall take place following origin acceptance and prior to machine teardown at OEM facility. Training manuals shall be provided for each student. The maintenance course shall cover at a minimum:

- a. I/O Board Diagnostics and Addresses
- b. Root Cause Analysis Troubleshooting
- c. Machine Routine Repair, Replacement, Realignment Procedures
- d. Add additional info provided by vendor

4.0 QUALITY ASSURANCE.

4.1 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection requirements specified herein and in accordance with the attached Quality Assurance Provision (QAP). Except as otherwise specified in the contract or purchase order, the contractor may use his own facility for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to witness any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items must meet all requirements of sections 3 and 4.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

- a. Contract quality assurance inspection (Origin) (see 4.4).
- b. Final Inspection/Acceptance test (Destination) (see 4.5).

4.3 Inspection Conditions. All inspections, tests, and examinations shall be performed in an indoor facility with ambient temperature conditions in the range of 55° F to 100° F and 10% to 95% relative humidity.

4.4 Contract Quality Assurance Inspection (Origin). Contract quality assurance inspection shall be applied at the contractor's facility prior to being offered for acceptance under the contract. The machine and all peripheral equipment added to the machine shall be fully assembled and perform all functions in accordance with this specification. Quality conformance inspection shall consist of the examination in 4.6 and the tests in 4.7. The

machine shall pass the examination, all tests, and contract quality assurance inspection to be acceptable for shipment.

4.5 Final Inspection/Acceptance Test (Destination). Final Inspection and acceptance test shall be performed on the machine to ensure conformance with this purchase description. The acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.6 Examination. The equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification.

4.7 Tests. Unless otherwise specified within this document, all programs, instruments, materials, material handling equipment, qualified personnel, and tools required to perform and evaluate the tests shall be furnished by the contractor. All measurement tools and indicating devices shall be calibrated in accordance with the attached QAP.

4.7.1 Alignment and Accuracy Tests. Tests shall be conducted to ensure the machine meets the alignment and accuracy tolerances specified in Table II. The machine repeatability and accuracy shall be checked.

Table II	
Positioning Repeatability	± 0.0002 inches
Positioning Accuracy	± 0.0001 inches

4.7.2 Operational Test. The machine shall be operated under CNC at no-load for not less than four (4) hours. Proper operation of all controls, motors, adjusting mechanisms. During operational testing the control functions and machine response to commands for slide motions, turret indexing, spindle speed changes, feed rate changes and auxiliary functions shall be checked. The machine shall be run in each mode of operation (Manual and Automatic) to test the machine's response to command data. Three (3) failures in a row to complete the run test may be cause for rejection of the entire system. After successful completion of the operational test, over-travel limits (programmed stops and hard stops), emergency stop devices, and accessories shall be verified during the trial period. The safety limit stops shall be checked by attempting to exceed the limits of travel of each moving axis.

4.7.3 Performance Test. The machine shall be tested in accordance with ISO 14137. The machine shall also be tested in accordance with the agreed upon ATP referenced in paragraph 3.8.4 utilizing the test part referenced in paragraph 3.8.5

Finished Workpiece Accuracy	± 0.0002 inches
Finished Workpiece Repeatability	± 0.0002 inches

4.7.4 Audible Noise Level Test. The equipment shall be tested for compliance with paragraph 3.1.4. “Audible Noise Level”.

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INSTALLATION RESPONSIBILITIES

Government	Contractor	
☐	☒	a. Provide machine foundation in accordance with the Manufacturer's design requirements
☐	☒	b. Furnish labor and material handling equipment for off-loading and placing item on foundation
☐	☒	c. Provide and install anchor bolts and nuts
☐	☒	d. Set and rough level the machine on its foundation
☐	☒	e. Level and align machine
☐	☒	f. Connect machine to the provided utilities hook-up
☐	☒	g. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests
☐	☒	h. Provide and charge all systems with fluids in accordance with manufacturer's instructions
☐	☒	i. Material for performance testing at Government facility
☐	☒	j. Material for performance testing at Contractor's facility
☐	☒	k. Material handling, Rigging equipment and Qualified Personnel for performance testing

GOVERNMENT RESPONSIBILITIES

- ☒ Provide utilities hook-up within 50 feet of the machine
- ☐ Government will connect utilities to the machine based on the contractor's accepted site drawings.
- ☐ Assign one mechanic (mechanical/hydraulic) and one maintenance mechanic (electrical/electronic) to provide information and any additional materials for the contractor's representative during installation, verification, and initial start-up of the machine.
- ☐ No work shall be performed by customer personnel when the contractor's representative is not in the immediate work area during installation, verification, and initial start-up of the machine.